

2(a)(i)	<i>Vibrio cholerae</i> ;	1
2(a)(ii)	A 1 <i>cell structure</i>: ribosome ; R RER 2 <i>difference</i>: 70S / smaller / 18 nm v 80S / larger / 25–30 nm ; B 3 <i>cell structure</i>: DNA / chromosome ; I RNA 4 <i>difference</i>: circular / (closed) loop v linear OR no histone proteins / naked v histone proteins OR not surrounded by nuclear envelope v surrounded by nuclear envelope ; A in a nucleus v not in a nucleus C 5 <i>cell structure</i>: cell wall ; 6 <i>difference</i>: murein / peptidoglycan v cellulose ; I lignin	6

Question	Answer	Marks
2(b)	<i>two from</i> 1 caused by, a pathogen / a bacterium / <i>V. cholerae</i> ; 2 transmissible / AW OR reference to faecal-oral route ; 3 reference to reduced effectiveness of functions / AW ;	2
2(c)	primary, secondary, tertiary ; A 1°, 2°, 3° quaternary ; A 4°	2
2(d)	<i>three from:</i> 1 cholera toxin, fits into / complementary to, receptor / GM1 ; A complementary shape 2 membrane pinches in / invaginates / AW ; A engulfs / envelops 3 membrane fusion ; 4 (endocytotic) vesicle / vacuole, formed ; 5 ATP / energy, required ; A points from an annotated diagram	3
2(e)(i)	<i>one from:</i> 1 portion that binds to cell ; 2 (antibodies produced) prevent binding to cell / prevent entry to cell ; 3 safer as not the toxic portion ; 4 A subunit, causes damage to cell / less safe / AW ; 5 AVP e.g. larger so more likely to provoke immune response / AW ;	1

Question	Answer	Marks
2(e)(ii)	<i>five from:</i> 1 vaccine contains (subunit B / bacterial) antigen(s) ; 2 primary immune response occurs ; 3 correct ref to B-lymphocytes / formation of plasma cells ; A B cells 4 secretion of, antibody / immunoglobulin (against cholera antigens) / antitoxins ; 5 T-helper lymphocytes secrete cytokine ; 6 (cytokine) increases humoral response / stimulates T-killer cells / stimulates macrophages ; 7 memory cell production ; 8 secondary (immune) response / response on further infection, is faster ; 9 higher levels of antibodies produced (during further infection) ; 10 active artificial immunity (against cholera) ; 11 AVP e.g. <i>idea</i> of specific antibody against each of the different vaccine antigens ;	5